

From path-dependent industrial policy to opportunity-cost accounting

CCS, UK ETS prices and alternative abatement in the Peak Cluster case

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TARGET JOURNALS

Primary: Energy Research & Social Science — mixed-methods energy-policy case, not process engineering. Secondary: Climate Policy, if the manuscript is cut toward instrument design (ICC CfD, cluster sequencing, diversification clauses). Reserve: Environmental Innovation and Societal Transitions, if the theoretical claim is placed in the foreground and the TEA moved to an annex. Nature Energy and Applied Energy are not appropriate: the Monte Carlo is literature-parameterised, not a new capture-plant model.

Working length: 7,000–8,000 words (ERSS research article). Authorship: Alhassan Abdulai (sole), with optional co-authorship by Prof. Kane.

WHAT IS NOT ALREADY IN THE GREY LITERATURE

- Separates PRTR-reported 2.248 Mt/yr from ICC design 3 Mt/yr in a live UK cluster.
- Places full-chain MEA LCO₂A (median USD 131.4/t) against UK carbon-price markers and reports the implied 15-year ICC bill at design throughput.
- Shows that institutional lock-in (mean 4.4/7; Peak Cluster materials = 7) co-occurs with an SLO distribution that has no approval cell — lock-in without legitimacy.
- Treats Development Consent Order as the point where path dependence and path change are still both possible.

EXTENDED ABSTRACT

The United Kingdom has organised industrial carbon capture around cluster sequencing and the Industrial Carbon Capture business model, a contract-for-difference that pays the gap between a negotiated strike price and the UK ETS reference. Peak Cluster is an early test of that architecture in a hard-to-abate minerals sector. This article asks whether that architecture is an efficient use of public abatement expenditure, and whether it can proceed without a social licence.

A convergent mixed-methods case study combines (i) lock-in coding of fifteen policy and industry documents, (ii) 10,000-iteration Monte Carlo simulation of full-chain LCO₂A for five capture options, tornado sensitivity and a sectoral MAC curve, and (iii) frame and SLO coding of 191 records. Subsidy cases use the 3 Mt/yr ICC design throughput, not the PRTR total of 2.248 Mt/yr.

Mean lock-in is 4.4 / 7. Post-combustion MEA has a median LCO₂A of USD 131.4/t (P5–P95 113.8–149.7) against a 2026 UK ETS civil-penalty reference of USD 65.2/t. The implied subsidy is about USD 66.2/t, or ~USD 199 million per year at design throughput if the low price path holds. Efficiency, LC3/SCM substitution and alternative fuels occupy the left of the MAC (mid-points USD 6.6, 33.0 and 37.0/t). Public discourse does not license the project: 51% scrutiny, 49% withholding; frames of place identity (49%) and procedure (34%).

The joint finding is a generative mechanism. Instruments that make CCS contractable also read, to affected communities, as procedural pre-determination. Cement CCS is not shown to be unnecessary; it is shown to

be premature as an exclusive 15-year public commitment under present prices and present licence conditions.

MANUSCRIPT PLAN

SECTION	LENGTH
Introduction — cluster sequencing, ICC CfD, three questions	~900 words
Framework — Unruh/Erickson/Seto lock-in; SLO; opportunity cost	~1,100
Methods — document scores; MC LCO ₂ A; MAC; discourse; integration	~1,200
Results — policy heatmap, TEA, SLO (no approval cell)	~3,300
Integration, discussion, conclusion	~1,600

RISKS AND TIMELINE

- **Dual publication.** Rewrite for the journal; do not copy the dissertation chapter-for-chapter.
- **Ethics.** Aggregate the social-media corpus; do not reproduce identifiable posts.
- **Desk-reject risk.** If TEA is presented as a FEED-quality cost model. Mitigated by stating literature-parameter Monte Carlo in the first methods paragraph.
- **Timeline.** Recast October–November 2026; ERSS submission targeted Q4 2026 / Q1 2027.